Answers: Identifying Population, Sample, and Sample Size

In every problem below, the number contacted and the number who actually responded are both given. The **sample** is the group that actually provided data, so the **sample size** is the number who responded — not the number invited.

1. Clothing Store Survey

a) **Population:** all customers of the clothing retailer.

b) **Sample:** the 684 customers who responded to the survey.

c) **Sample size:** $n = 684$.

The 2,500 customers who were emailed are the group contacted, not the sample. Response rate: $684/2500 = 27.4\%$.

2. College Students

a) **Population:** all 8,200 undergraduate students at the college.

b) **Sample:** the 427 students who returned completed questionnaires.

c) **Sample size:** $n = 427$.

600 students were selected; 427 responded. Response rate: $427/600 = 71.2\%$.

3. Supermarket Customers

a) **Population:** all customers of the supermarket chain (all 74 stores).

b) **Sample:** the 1,054 customers who answered the question.

c) **Sample size:** $n = 1{,}054$.

1,200 customers were approached and 146 declined. Response rate: $1054/1200 = 87.8\%$. Note also that customers were approached at only 15 of the 74 stores, so the sample does not cover the whole chain evenly.

4. Streaming Service

a) **Population:** all approximately 12 million subscribers of the streaming service.

b) **Sample:** the 2,318 subscribers who completed the survey.

c) **Sample size:** $n = 2{,}318$.

10,000 invitations were sent. Response rate: $2318/10000 = 23.2\%$.

5. Department Stores

a) **Population:** all approximately 4,600 department-store locations in the United States.

b) **Sample:** the 218 stores whose managers provided sales information.

c) **Sample size:** $n = 218$.

300 stores were contacted. Response rate: $218/300 = 72.7\%$.